CV PROJECTS | COMPLETION GUIDE FOR MITACS INTERVIEWS

# Finish Your CV Projects

A step-by-step guide to completing every incomplete project on your CV, with the software downloads, the finished proof for each one, and the honest answer to give when a professor asks about it.

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Built for Mitacs Globalink professor matching — and for any professor who reads your CV and asks, "Show me."

## What this guide promises — and what it does not

**It promises:** a finish line for each project, in order, with the exact steps, the software to download, and a "done test" you can check with your own eyes. When you pass the done test, you can say the project is finished. Before that, you keep the honest wording this guide gives you.

**It does not promise:** that a professor will select you, that a trial licence will still be free, or that a project can be finished by reading about it. Each chapter ends in a file, a screenshot, or a running model. No file, no finish.

> **Read this first — FYP first until 6 November.** Your second FYP presentation is Friday 6 November 2026. Do not start HOMER, Energiscope, or any new software work before that presentation is delivered. This guide is ordered so the fastest credibility fix (Energiscope, under an hour) comes right after the presentation, and the longer builds come after it. If you are reading this before 6 November, read only Chapter 1 and Chapter 2, then put it down.

**How to use this guide:** work through it in chapter order, one project at a time. Do not start five projects at once. At the end of each chapter, tick the done test, save the proof in the folder structure in Chapter 12, and only then update your CV wording to match.

Prepared 4 November 2026, from your refined CV (2 pages, delivered October 2026), your project files already on hand, and official software pages checked in November 2026. Licence prices, trial lengths, and download pages change — re-check the official page on the day you download. Never download MATLAB, HOMER Pro, or any paid software from a cracked or "free full version" site: a pirated licence is a reason for a university to drop you, and a professor will not defend it.

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### The golden rules for every project in this guide

1. **Show, don't claim.** A professor believes a screenshot, a file, and a number you can explain. "I built a model" with nothing to open is worse than "I am building it, here is the working part."
2. **Never upgrade a status in speech.** "In preparation" is not "submitted". "Sample data" is not "real NEPRA data". "In progress" is not "completed". This guide keeps those words apart on purpose.
3. **One finish at a time.** Finish Energiscope's honesty fix completely before you open HOMER. A half-finished HOMER case plus a half-fixed website is two new questions, not two answers.
4. **Keep the originals.** Never overwrite your only copy of a dataset, workbook, or model. Copy it, work on the copy, and keep the original folder untouched. This rule is in every chapter because it is the one that saves you when a step goes wrong.
5. **Write the number down with its source.** Every power figure, cost, and efficiency in your final files carries its input and its source in the same file. A number with no source is a question you cannot answer.

> **Your machine, as this guide assumes it:** Linux Mint 22.2 Xfce, Intel Core i3, about 8 GB RAM, about 81 GB disk. That matters twice: HOMER Pro does **not** run on Linux (Chapter 3 gives you the Windows routes), and large DEM and video files must not be copied onto the small laptop disk without checking free space first. Where a step needs Windows, the guide says so plainly instead of pretending otherwise.

## 1. Your CV, project by project

This chapter is taken from your refined 2-page CV, line by line. "CV says" is what a professor reads. "True status" is what you can actually show today. The gap between the two columns is the whole job of this guide.

| Project (CV order) | CV says | True status today | What a professor will ask |
|---|---|---|---|
| **FYP — low-head hydropower, KP canals** | 12 sites, data from 3 departments, techno-economic comparison, suitability ranking "not yet complete" | Presentation 6 Nov 2026; Warsak data requested, not yet received as a file; ranking not built on real site data; GIS map in progress; no site power figures invented | "Show me one site's head and flow, and how you ranked it. What did the data actually say?" |
| **Energiscope — Pakistan energy data platform** | "Real NEPRA data, 10+ years, ~40 GW" | **Do not say this as it stands.** The reviewed repo holds 6 years of sample data (2020–2025) hardcoded in app.js, not a 10-year NEPRA dataset file. Honesty fix files are ready but **not uploaded**; the live site still shows the old wording | "Where did the data come from? Can I download the dataset? Why does this year differ from NEPRA's own table?" |
| **Hybrid system — Gulabad, Charsadda (HOMER Pro)** | "In progress"; preliminary load profile and solar resource done | Load and solar resource are preliminary only. No finished HOMER sizing, no cost result, no report. HOMER Pro is Windows software; your laptop runs Linux Mint | "What load did you serve, what did HOMER optimise, and what did it cost? Show me the HOMER file." |
| **Floating PV research** | 120 weekdays of field data, Excel workbook, "data collection and analysis complete; manuscript in preparation" | Workbook exists; the manuscript is not a draft a professor can read yet. Your exact contribution and the headline result are not written down in one place | "What was your contribution? What did floating beat ground-mounted by, and how did you calculate it?" |

## 1 (continued). Your CV, project by project

| Project | CV says | True status today | What a professor will ask |
|---|---|---|---|
| **Tarbela 5th Extension — project intern (PCCCL, hybrid)** | Sept 2026 – present, 3-month placement, installation and rewiring exposure | Ongoing. No written logbook, certificate, or one-page summary exists yet to show what you personally did | "What did you personally install, wire, or test? Who supervised you?" |
| **WAPDA Tarbela — 4-week intern** | Turbines, generators, switchyard, high-voltage, load coordination | Completed as experience. Needs only an evidence sheet (dates, units visited, tasks) so you can answer precisely instead of generally | "Which unit, which switchyard task, what did you operate or only observe?" |
| **Skills: Python (developing), MATLAB (working), HOMER Pro (in progress), GIS** | Listed as skills | Python Lesson 01 done; its independent task (Site F, capacity factor, 500 kW head) is pending. MATLAB PV model is stuck on the Simscape Electrical check. QGIS DEM clip is unfinished (corridor polygon being drawn, no clipped raster confirmed). CanalPower PK is a starter pack, not yet a GitHub repo | "Run it. Show me the code, the map, and the PV curve — now, not next week." |
| **Coursework models** (wind turbine prototype, DC-DC boost converter, sand thermal storage, solar electrolyzer, fire suppression, solar water heater) | Designed / built / modelled | Finished as coursework. No consolidated evidence pack: no single sheet per model with inputs, results, and a photo or model file | "Pick one. What were the inputs, what came out, and what would you change?" |

**The three answers that damage you if you give them unprepared:** (1) repeating the "10+ years of real NEPRA data" line when the repo is sample data; (2) describing the FYP ranking as trained or validated when it is physics-first with a model as a comparative check only; (3) calling the floating-PV manuscript "published" or "submitted" when it is in preparation. Chapter 11 gives you the safe, true version of each answer. Learn those three first.

## 2. The order to do them in

Do not work in CV order. Work in risk order: fix what can embarrass you first, then finish what a professor will ask about most, then polish the rest.

| Order | Project | Why this early | Time (honest estimate) |
|---|---|---|---|
| 0 | **FYP presentation, 6 Nov** | It is this week. Nothing in this guide outranks it | Until 6 Nov |
| 1 | **Energiscope honesty fix (P0)** | Fastest credibility repair; removes the one claim a professor can disprove in two minutes | 30–60 minutes |
| 2 | **Interview answer pack (Chapter 11)** | You can be asked before any project is finished; this costs no software | One evening |
| 3 | **FYP ranking on real data** | Your main project; it is also the story every other answer points back to | 2–3 weeks after data arrives |
| 4 | **HOMER Gulabad case** | Needs a Windows machine and a 21-day trial — start the trial only when the load and resource data are ready (Chapter 6) | 2–3 weeks, inside one trial if you plan it |
| 5 | **Floating PV manuscript pack** | Needs your supervisor's eye; start the figures now, send the draft when it is readable, not perfect | 2–4 weeks, supervisor time included |
| 6 | **Skills proof (Python, QGIS, MATLAB PV)** | Small, checkable wins that back up the skills line on your CV | A few evenings, one skill at a time |
| 7 | **Tarbela record + coursework sheets** | Writing only; do it last because it needs no software and no one else's data | One weekend |

**A realistic calendar.** Week of 6 Nov: presentation only. 7–9 Nov: Energiscope P0 upload plus the interview pack. 10–30 Nov: FYP ranking and HOMER in parallel only if the Windows machine is already arranged — otherwise HOMER after the FYP ranking. December: floating-PV draft to your supervisor, skills proof, and the Tarbela record. If Mitacs matching moves faster than this, Chapters 4 and 11 alone already remove the worst risks; tell a professor the rest is in progress with dates, and show the dated plan in Chapter 12.

> **What "finished" means in this guide.** A project is finished when (a) its output file opens, (b) its inputs and sources are written in the same file or folder, (c) you can explain its main result in two minutes without notes, and (d) the CV wording matches exactly what the file shows. If any one of those four fails, the project is still in progress — say so.

## 3. Software downloads and installation

Download only from the official site in the table. "Free download" sites that bundle installers are how laptops pick up malware and how cracked licences appear in your history. If your university already licences a tool, use that licence first — ask your department or lab in charge before you pay for anything or start a trial.

| Software | What it is for in this guide | Official source | Cost / licence note (check on the day) |
|---|---|---|---|
| Python 3 | FYP power/ranking code, CanalPower, Lesson 01 task | python.org, or your Linux package manager | Free, open source |
| Visual Studio Code | Editing Python code comfortably | code.visualstudio.com | Free |
| Git | Uploading Energiscope fixes and CanalPower to GitHub | git-scm.com, or apt on Mint | Free, open source |
| QGIS | DEM clipping, canal corridor map, FYP site map | qgis.org | Free, open source |
| MATLAB + Simulink + Simscape Electrical | PV model, DC-DC converter evidence, any Simulink proof | mathworks.com (your existing R2025b licence first) | Paid / campus licence; Simscape Electrical is a separate licensed product |
| HOMER Pro | Gulabad hybrid sizing | homerenergy.com (download from users.homerenergy.com after signing in) | **Windows only.** First-time users get a 21-day free trial with all modules (HOMER's own FAQ). Paid licence after that; check academic pricing on the site |
| LibreOffice Calc | Floating-PV workbook if Excel is unavailable on Mint | libreoffice.org, or apt on Mint | Free, open source |
| NASA POWER data | Solar resource for HOMER and PV work | power.larc.nasa.gov (web download, no install) | Free data access |
| GitHub + Vercel | Hosting Energiscope and CanalPower | github.com, vercel.com | Free tiers exist; your Energiscope repo is already on GitHub and deployed on Vercel |

**Before you install anything:** (1) check free disk space — on an 81 GB disk, do not download three large installers at once; (2) plug in power and use a stable connection, because a half-downloaded MATLAB or QGIS install wastes an evening; (3) write down the version you installed and the date, in the project folder, so a professor's "which version?" has an answer.

## 3 (continued). Install Python, VS Code, and Git on Linux Mint

Do this first. It is free, it is fast, and every later chapter assumes it works.

1. Open a terminal. Update the package list: `sudo apt update`
2. Install Python and the tools around it: `sudo apt install python3 python3-pip python3-venv git`
3. Check it worked: `python3 --version` and `git --version`. Both should print a version number. If either says "command not found", stop and fix that before going on — later steps will fail silently without it.
4. Make one project folder and one virtual environment inside it, so packages for one project never break another:

```
mkdir -p ~/projects/canalpower
cd ~/projects/canalpower
python3 -m venv venv
source venv/bin/activate
```

5. Install VS Code from code.visualstudio.com (download the .deb for Debian/Ubuntu, double-click it in the file manager, install), or from the terminal with the package your Mint version offers. Open the project folder in VS Code with `code .` from inside the folder.
6. Set your Git identity once, so commits carry your name:

```
git config --global user.name "Muhammad Ismail"
git config --global user.email "muhammadismail.khan.1047@gmail.com"
```

7. Test the whole chain in five minutes: write a file `test_power.py` containing the FYP formula P = 1000 × 9.81 × Q × H × η, run it with a known case (Q = 2.0, H = 3.0, η = 0.70 must give 41.2 kW), and confirm the answer. If that test passes, your Python setup is done. If it does not, the problem is the setup, not the physics — fix the setup now.

> **Done test — Python:** `python3 --version` prints a version, the 41.2 kW test passes, and you have a folder `~/projects/` with a working virtual environment. Save a screenshot of the terminal output in the project folder.

## 3 (continued). Install QGIS on Linux Mint

QGIS is free, but on Mint the version in the default software list can be old. The official QGIS route below is the one QGIS itself documents for Debian/Ubuntu-based systems, including Mint. Mint's codename is not its own name: QGIS's guide notes that on Mint, `. /etc/os-release; echo "$UBUNTU_CODENAME"` shows the Ubuntu base codename to use.

1. Find your base codename first: `. /etc/os-release; echo "$UBUNTU_CODENAME"`. Write it down — you need it in step 3.
2. Install the key tools and the QGIS signing key:

```
sudo apt install gnupg software-properties-common
sudo mkdir -m755 -p /etc/apt/keyrings
sudo wget -O /etc/apt/keyrings/qgis-archive-keyring.gpg
https://download.qgis.org/downloads/qgis-archive-keyring.gpg
```

3. Create the QGIS source file `/etc/apt/sources.list.d/qgis.sources` (with `sudo` and a text editor) containing, with your codename in place of the placeholder:

```
Types: deb deb-src
URIs: https://qgis.org/debian
Suites: YOUR-UBUNTU-CODENAME
Architectures: amd64
Components: main
Signed-By: /etc/apt/keyrings/qgis-archive-keyring.gpg
```

4. Install: `sudo apt update` then `sudo apt install qgis qgis-plugin-grass`
5. Open QGIS, go to Help → About, and note the version. Then load one of your Copernicus DEM tiles (Chapter 9) and confirm it draws. If QGIS will not start or a tile will not draw, do not start the FYP map — fix the install first.

**Fallback if the repository route fails:** install QGIS as a Flatpak from the Software Manager (search "QGIS", install the org.qgis.qgis package). Flatpak bundles its own libraries, which is why it often works when the apt route hits dependency trouble. The trade-off is larger downloads and a sandboxed file view — you must grant it access to your project folder before it can see your DEM files.

> **Done test — QGIS:** QGIS opens, its version is written in your notes, and a Copernicus DEM tile displays with a correct-looking elevation range. Screenshot the About box and the loaded tile.

## 3 (continued). MATLAB and Simscape Electrical — check first, install second

You already have MATLAB R2025b. The open question from your earlier work is not "where is MATLAB?" but "does your licence include Simscape Electrical?" There is no separate PV library to download: the PV Array block lives in Simscape Electrical (Specialized Power Systems). Do the checks in order — they take ten minutes and decide which path below you follow.

1. Open MATLAB. Run `ver` and look for "Simscape Electrical" in the list. Also run `license('test', 'Power_System_Blocks')`: 1 means you have the licence, 0 means you do not.
2. Open the Library Browser and search "PV Array". If the block appears, your licence and install are fine — skip to Chapter 9 and build the model.
3. **If the block is missing but the licence test is 1:** install the product you already own. In MATLAB, go to Home → Add-Ons → Get Add-Ons, search "Simscape Electrical", and install it. Alternatively, re-run the MathWorks installer for R2025b and tick Simscape Electrical (and Simscape, which it needs) in the product list.
4. **If the licence test is 0:** you do not own it yet. Check first whether UET has a campus licence that covers you, or whether your MATLAB licence is a Student Suite (MathWorks' own answer notes Simscape Electrical is included in the current Student Suite). Only then consider buying it. Do not download it from any other site.
5. **Fresh MATLAB install (only if MATLAB is not on the machine you are using):** sign in at mathworks.com, go to Downloads, choose R2025b for Linux, download the installer, unzip it, run `./install` from a terminal, sign in with your MathWorks account, select your licence, tick MATLAB, Simulink, Simscape, and Simscape Electrical, and let it finish on power with a stable connection. It is a multi-gigabyte download — do not start it on a weak connection the night before you need it.

> **Common trap:** downloading a "Simscape" file from a forum or file-sharing link and trying to link it by hand. Simscape Multibody Link files for SolidWorks are Windows-only and are not Simscape Electrical. If a download asks you to run a Windows-only linking script on Linux, you have the wrong file — stop.

> **Done test — MATLAB:** either the PV Array block opens in the Library Browser, or you have a written note saying which route you are on (licence owned / campus licence requested / Student Suite to buy) plus a working plain-Simulink single-diode fallback model from Chapter 9. "I will check later" is not a done test.

## 3 (continued). HOMER Pro — Windows only, and the 21-day clock

This is the step most students get wrong, so read it before you download anything. HOMER Pro and HOMER Grid are **Windows-based applications** (HOMER's own FAQ). They do not run natively on Linux Mint. Your options, in order of preference, are:

1. **A Windows PC you already have access to** — a UET lab machine, a department computer, or a borrowed Windows laptop. This is the cleanest route: install there, work there, and keep the .homer project file on a USB drive and in your own backup.
2. **A Windows virtual machine on your Mint laptop** — install VirtualBox from the Software Manager, create a Windows virtual machine, and install Windows inside it (you need a legitimate Windows licence or your university's education licence). This works but is slow on 8 GB RAM: give the virtual machine at least 4 GB, close everything else while HOMER runs, and expect longer optimisation runs.
3. **Do not** try to force HOMER through Wine or an unsupported emulator and then present the result as a normal HOMER run. HOMER itself says Mac/Windows-emulator installs are unsupported, and a crashed optimisation in the middle of your case is not a result.

**Download and trial (on the Windows machine):** go to homerenergy.com, click a "21-Day Free Trial" link or Sign In, create an account, confirm your email, complete the profile questions, read and agree to the HOMER End User Agreement, and click "Download HOMER Pro". Run the downloaded `HomerPro-X.X.X.msi` file (choose the 64-bit version for a 64-bit Windows). First-time users get a fully functional 21-day trial with all modules.

> **Start the trial only when you are ready to run.** The 21 days start at activation, not at your first serious run. Do not activate the trial the day you download it "just to look". First finish Chapter 6's load profile and NASA POWER resource download, so that day 1 of the trial is the day you build the system — otherwise the trial expires halfway through your sensitivity runs and you are stuck mid-case.

**NASA POWER (no install):** open power.larc.nasa.gov, use the Data Access Viewer, enter Gulabad's coordinates, select solar irradiance (and temperature/wind if you need them), choose your date range, and download the CSV. Save the CSV with its source, coordinates, and download date written in the filename or in a notes file — a resource file with no coordinates is useless to a professor.

> **Done test — HOMER setup:** you can name the Windows machine you will use, HOMER Pro opens on it, and your load CSV and NASA POWER CSV are saved and dated. Do not count "downloaded the installer" as done.

## 4. Project 1: Energiscope — fix the honesty problem first

**Why this is first:** every other project can be explained as "in progress". This one cannot, because the live site currently says something the repo does not support. A professor who opens the site and then opens the repo will find the gap in minutes. The fix below is already prepared for you — it only needs uploading.

### Step A — the P0 upload (30–60 minutes)

1. On your own machine, make a backup copy of the repo as it is today. Never edit the only copy.
2. Unzip `Energiscope_P0_Fixes.zip` (prepared 4 November 2026). It contains replacement files: `index.html, app.js, styles.css, admin.html, admin.js, README.md`.
3. Open your GitHub repo `engr-energy/pakistan-energy-dashboard` in a browser. For each file in the zip, open the matching file in the repo, choose Edit (pencil icon), replace the whole contents with the zip version, and commit with a clear message such as "P0 honesty fix: label sample data, remove real-time claim". Do this file by file — do not upload the zip itself as a repo file.
4. Wait for Vercel to redeploy (it redeploys automatically from GitHub). Then open the live site in a fresh browser tab and check, with your own eyes: (a) the words "real-time" are gone, (b) the data is labelled illustrative/sample, (c) the Data Note banner shows, (d) the admin area is labelled "Admin (local demo)". Screenshot all four.
5. Update your CV the same day: change "10+ years of real NEPRA data, ~40 GW" to "sample generation data (2020–2025) visualised; real NEPRA dataset integration in progress — see Chapter 4, Step B". Until the live site shows the fix, keep the old CV wording out of any new application you send.

> **Done test — P0:** the live site (not just your local copy) shows the corrected wording and the Data Note banner, and you have four screenshots dated today. If the live site still shows the old wording, the job is not done — check whether Vercel redeployed and whether you edited the repo branch that Vercel deploys from.

### Step B — the full fix (after P0, before you call Energiscope "finished")

1. **Remove the hardcoded admin password** from client-side JavaScript and from the README. A password printed in public code is not a demo feature; it is a security fault. Replace the admin area with a clearly labelled local demo, or remove it until you can protect it properly. Never paste the password into a chat, email, or application to "prove" anything.
2. **Fix the broken sample-download links** (they point to `samples/electricity.csv` while the files are named `*_sample.csv`) or remove the links until the files exist at the linked paths.
3. **Choose one canonical dataset and say so.** The repo currently holds three disagreeing datasets. Pick one, delete or clearly archive the others, and write a Methods page stating: source, years, unit (TWh, not the old mixed unit), what "sample" means, and what is not yet verified. Only after a real, sourced NEPRA dataset is actually loaded may the CV say "NEPRA data" again — and then it must say the exact years and source.
4. **Add the promised pages one at a time:** Dashboard, Data & Downloads, Methods, Reports, Articles, Learn, About — in that order, with the data-status section on each page that shows data.

## 5. Project 2: Your FYP site-suitability ranking

This is your main project, and it is the one a Mitacs professor in hydropower will spend the most time on. The ranking is **physics-first**: power from head, flow, and efficiency ranks the sites, and any model is only a comparative check on top of that. Never present it the other way round, and never present a ranking built on guessed heads as a measured result.

### Step 1 — get the data into a file you can defend

1. When Warsak/Irrigation data arrives, save the original file untouched, with the sender, date, and office name in the filename or a receipt log. Log "requested" vs "received" separately — a promised file is not a received file.
2. Build one spreadsheet, one row per site, with these columns and no others added without a source: site name, canal, structure/fall, chainage if given, season, gross head (m), net head (m) with losses stated, flow ($m^3/s$), source of each number (department letter, field measurement, or "not yet sourced"), and date.
3. Flag every cell that is still unsourced in red. A ranking that hides its unsourced cells is a ranking a professor will not trust; a ranking that shows them is a ranking you can defend while data is still arriving.
4. Photograph or scan the nameplate/project board of the visited small plant (the 30 kW vs 100 kW question). Until that photo exists, keep the two figures separate in every document: "30 kW reported at the visited plant" and "100 kW DoST inventory entry, same installation unproven". Do not merge them.

### Step 2 — calculate power, transparently

1. For each site and season, calculate $P = \rho g Q H \eta$ with $\rho = 1000$ kg/m³, $g = 9.81$ m/s², $Q$ = seasonal flow, $H$ = net head, $\eta$ = stated efficiency with its source. Show the formula, the inputs, and the result in the same row. Use the tested Python module you already have (`power_calculator.py`, and the CanalPower `power.py` module) rather than a new spreadsheet formula you cannot test.
2. Run the worked check first: 2.0 m³/s × 3.0 m × 0.70 must give 41.2 kW. If your sheet gives anything else, fix the sheet before you trust any site result.
3. Calculate annual energy from seasonal flows, not from one flow times 8,760 hours. State the operating hours or capacity factor you used, and why. A single-flow annual figure is the first thing an examiner will challenge.
4. Apply the irrigation veto before ranking: any site that would harm irrigation timing, or lacks the needed consent, is marked "not feasible on irrigation grounds" no matter how much power it shows. Power without the veto is not a feasibility result.

## 5 (continued). FYP ranking — ranking, economics, and the model check

### Step 3 — rank, then check with the model (in that order)

1. Rank sites by physics first: net power at seasonal low flow, then annual energy, then economics. Write the ranking rule at the top of the ranking sheet so anyone can re-derive it.
2. Only then run the comparative model check (Random Forest suitability/power, Isolation Forest anomaly monitoring, as in your roadmap). State its limits in the same document: small sample (12 sites), no real turbine performance data, and label-circularity risk if the model is trained on labels derived from the same physics it is meant to check. Deep learning is out of scope — do not add it to impress anyone.
3. If the model ranking and the physics ranking disagree on a site, report the disagreement and investigate the site — do not silently pick the prettier ranking.
4. Settle the head-threshold question with your supervisor before you finalise: the 2 m figure is currently a discussion point only, not a changed rule. Record the supervisor's decision and date in the ranking file. Do not change the threshold in any deck or script on your own.

### Step 4 — economics and the final pack

1. For feasible sites only, add: estimated cost (with source or "estimate" label), LCOE, NPV, IRR, and payback, each with its assumptions (discount rate, lifetime, tariff or avoided cost) written beside it. An economics table with no assumptions is not checkable.
2. Produce the FYP evidence pack: (a) the sourced data spreadsheet, (b) the power/ranking sheet, (c) the GIS site map (Chapter 9), (d) captioned field photos, (e) the measurement method note, (f) a one-page limitations sheet. That pack is what you open when a professor says "show me".
3. Update the CV only when the pack exists: "site-suitability ranking completed on sourced seasonal heads and flows for [number] sites; physics-first ranking with model as comparative check; techno-economic comparison for feasible sites." If only some sites have sourced data, say exactly how many.

> **Done test — FYP ranking:** a professor can pick any ranked site and trace it back to a sourced head, a sourced flow, a power calculation, and an irrigation check, in under two minutes, from files you open on the spot.

## 6. Project 3: HOMER Pro — the Gulabad, Charsadda hybrid case

Your CV says: PV-hydro-battery hybrid for an unelectrified site in Charsadda, using HOMER Pro and NASA POWER data, with a preliminary load profile and solar resource assessment completed. The finish line is a HOMER project file, an optimisation result, and a short report — not "I used HOMER".

### Step 1 — finish the load profile before you touch HOMER

1. Write the load as a list, not a feeling: each load type (lighting, fans, phone charging, a pump, a small shop load — only what is real for Gulabad), its power (W), how many, and hours per day. Sum to daily energy (kWh/day) and peak (kW). Note the source of each item: household count, survey, or stated assumption.
2. Build the hourly shape: a 24-hour profile for a typical day, plus a seasonal note (summer vs winter) if the load changes. HOMER needs an hourly load; a single daily number is not a profile.
3. Save it as a CSV with hour, load (kW), and a notes column. This CSV is also your defence: when a professor asks "whose load is this?", you open this file, not a HOMER screenshot.
4. Decide the hydro input honestly: if you have a measured or department-sourced flow and head for the Gulabad site, use them; if you do not, model hydro as a stated scenario ("hydro at X $m^3/s$ and Y m head, to be confirmed by measurement") and label every hydro result as scenario-dependent. Do not present a scenario hydro plant as an existing resource.

### Step 2 — download the resource data

1. From NASA POWER (Chapter 3), download solar irradiance for Gulabad's coordinates, monthly averages at minimum, hourly if you plan an hourly run. Save the CSV with coordinates and download date.
2. Check the numbers for sense: annual average irradiance for Charsadda should be in a plausible range for northern Pakistan, and summer months should exceed winter months. If a month looks wrong, re-download before you model — a wrong resource file poisons every HOMER result.
3. Note the limits in your report: NASA POWER is satellite/modelled data, not a ground measurement at Gulabad. A professor respects that sentence; hiding it invites a harder question.

## 6 (continued). HOMER Pro — build, run, and report

### Step 3 — build the system in HOMER (only now start the trial)

1. On the Windows machine, start the 21-day trial (Chapter 3) and create a new project named `Gulabad_Hybrid_Charsadda`. Set the location to Gulabad's coordinates and load the NASA POWER solar resource.
2. Add components one at a time, with sizes you can justify: PV array (kW), hydro turbine (kW, with your head/flow scenario stated), battery bank (kWh, with type), converter/inverter (kW), and diesel generator only if your case genuinely includes one — your CV mentions both "PV-hydro-battery" and "PV-diesel-battery" in different places, so pick the case you actually modelled, name it consistently everywhere, and do not claim both unless you ran both.
3. Enter real costs for each component (capital, replacement, O&M) with a source or an "estimate, to be quoted" label, plus fuel price if diesel is included, and the project lifetime and discount rate. HOMER's economics are only as honest as these inputs.
4. Load your hourly load CSV. Check that HOMER's daily energy and peak match your spreadsheet. If they do not, fix the load file — do not adjust HOMER to force a match.

### Step 4 — run, read, and write it up

1. Run the optimisation. Save the winning system and at least two alternatives (for example, PV-battery without hydro, and PV-diesel) so you can say why the winner won.
2. Record, for the winning system: net present cost, cost of energy (per kWh), renewable fraction, unmet load, excess electricity, battery autonomy, and diesel hours/fuel if diesel is present. Copy these into your report exactly as HOMER shows them — do not round a cost of energy into a nicer number.
3. Run one sensitivity: fuel price or load growth, whichever matters more for Gulabad. One sensitivity, explained, beats five unexplained sensitivity tables.
4. Write a 3–4 page report: site and load, resource, components and costs, winning system with its numbers, alternatives, sensitivity, limitations (including the hydro scenario and the NASA POWER caveat), and what you would measure next. Save the .homer file with the report.

> **Done test — HOMER:** you can open the .homer file, show the winning system, read its cost of energy and renewable fraction aloud, and explain one input behind each number. Then, and only then, change the CV line from "in progress" to "techno-economic sizing completed for [system], cost of energy [value] per kWh (HOMER Pro, [date])".

## 7. Project 4: Floating PV — from workbook to manuscript draft

Your CV says: 120 weekdays of field data, a structured multi-sheet Excel workbook, data collection and analysis complete, manuscript in preparation. A professor will not ask you to prove 120 days from memory — they will ask what you found and what your part was. This chapter turns the workbook into answers.

### Step 1 — audit the workbook (one evening)

1. Copy the workbook; work on the copy. List every sheet, what it holds, and which sheets are raw data vs calculated results. Any sheet you cannot explain in one sentence gets a note at its top explaining it — or it gets moved to an appendix folder.
2. Check the three comparisons the CV promises: irradiance, temperature, and output efficiency, floating vs ground-mounted. For each, confirm: same days, same time window, same panel type or a stated difference, and a named calculation (for example, efficiency = output power ÷ (irradiance × panel area)). If any comparison mixes different days or different panels without saying so, fix the calculation and note the correction.
3. Produce one headline table: for floating and ground-mounted separately, mean irradiance, mean panel temperature, mean output, and mean efficiency, with the number of days behind each mean. That table is your interview answer in numbers.
4. Write your contribution down in three lines: what data you personally collected, what analysis you personally ran, and what someone else (supervisor, co-researcher) did. Professors ask "your contribution" to every multi-author project — a vague answer here hurts more than a modest true one helps.

### Step 2 — figures and draft

1. Make three figures from the audited workbook, nothing fancier: (a) daily efficiency, floating vs ground, over time; (b) temperature vs efficiency scatter; (c) monthly mean output comparison. Each figure gets a title, axes with units, and a one-line caption stating what it shows. A figure without units is not a figure.
2. Draft the manuscript in this skeleton, one section at a time: Title; Abstract (one paragraph: question, method, headline result, implication); Introduction (why floating PV matters, what is already known, the gap); Methods (site, panels, measurement, days, calculations, limits); Results (headline table + three figures, no interpretation yet); Discussion (what the result means, why temperature/efficiency behaved as it did, limits, what to measure next); Conclusion (three sentences); References (only papers you have actually read).
3. Send the draft to your supervisor with a short note: what you need from them (methods check? results check? authorship order?) and by when. "Manuscript in preparation" stays the correct CV wording until a journal submission exists — then it becomes "submitted to [journal], [date]", and only after acceptance does it become anything stronger.

> **Done test — Floating PV:** you can open the workbook, show the headline table, explain your personal contribution in three lines, and show a readable draft with figures. That is a finished "in preparation" project — it does not need a publication to be defensible.

## 8. Project 5: Tarbela internships — turn experience into a record

Internships are not finished by time passing. They are finished when you can state, precisely, what you did, where, under whom, and what you can now do because of it. Both Tarbela entries need the same one-page record.

### For the 5th Extension project internship (ongoing, Sept–Dec 2026)

1. Start a logbook now, not at the end: date, place (site/office/remote), task, what you personally did vs observed, equipment or drawing involved, supervisor present. One line per day you worked is enough; a blank week is a week you will not remember in an interview.
2. Collect, as you go: the placement letter or email, any task sheet or drawing you worked from (only if you are allowed to keep a copy — ask before taking site documents), and photos you are permitted to take. Never share restricted site drawings or internal documents with a professor or online; describe the work, do not leak the document.
3. At the end of the placement, ask for a certificate or completion letter stating dates, role, and supervisor. If a formal certificate is not issued, ask your supervisor for a short confirmation email you may quote. Save it with the logbook.
4. Write the one-page summary: placement, dates, hybrid pattern (which days on site, which remote), three tasks you did best (with a sentence each on what you did and what it taught you), and one limitation ("observed, not operated" where that is true). That page is what you attach or quote in an interview.

### For the WAPDA Tarbela 4-week internship (completed)

1. Reconstruct the same record from memory and any notes, photos, or certificate you still hold: weeks, units/areas visited (turbines, generators, switchyard, high-voltage, load coordination), and for each, "operated / assisted / observed" — honestly, separately.
2. If you no longer hold a certificate, do not invent one. Write "certificate: [held / not held / requested on (date)]" in the record. A professor checks consistency, not paper alone.
3. Keep both Tarbela records separate. They are different placements, different employers (WAPDA station vs PCCCL project), and merging them into one "Tarbela experience" story is exactly the kind of blur a careful interviewer will pull apart.

> **Done test — Internships:** for each placement you can state dates, supervisor, three personal tasks, and operated vs observed, from a one-page record, without hesitation or generalities.

## 9. Skills proof — Python, QGIS, and the MATLAB PV model

A skills line on a CV is a promise to demonstrate. Each skill below has a small, finished artefact. Do one at a time, in this order: Python first (it feeds the FYP), QGIS second (it feeds the FYP map), MATLAB PV third (it is the stuck one, and it has a fallback).

### Python — finish Lesson 01's independent task

1. Open your Lesson 01 code ($P = \rho g Q H \eta$, five synthetic sites). Add Site F: flow 10.0 m³/s, head 4.0 m, efficiency 0.81. Calculate and record its power.
2. Test a capacity factor of 0.45 on your sites: annual energy = power × 8,760 × 0.45. Write down what the factor means (the plant does not run at full power all year) and why 0.45 is a test value, not a measured one.
3. Work backwards: what head is needed for 500 kW at flow 12.0 m³/s and efficiency 0.82? Rearrange $H = P \div (\rho g Q \eta)$ and show the arithmetic. Check the answer by running it forwards.
4. Save the script, its output, and a one-paragraph note stating that all Lesson 01 numbers are illustrative, not FYP site data. That note is what stops a professor mistaking a classroom example for a site result.

### CanalPower PK — from starter pack to a repo you can link

1. Unzip `CanalPower_PK_Starter_Pack.zip` into a fresh folder. Run the tests first (`python -m pytest tests/` or the test runner in the README) and confirm they pass. A repo whose tests fail on first run is not a portfolio piece.
2. Run the Streamlit skeleton (`streamlit run app.py` — install Streamlit in your virtual environment first if needed) and screenshot it working with the five synthetic example sites.
3. Create a GitHub repo, push the code with the README and the honesty rules (never invent data; irrigation veto wins; tests before commits), and add the repo link to your CV only after the tests pass and the app runs. Until then it stays "starter pack", not "open-source tool".
4. Later, and only after the FYP ranking is done, add one real-data mode: the same power module reading your sourced FYP spreadsheet. That single step turns a synthetic demo into a tool tied to your actual project.

> **Done test — Python/CanalPower:** Site F's power, the 0.45 capacity-factor energies, and the 500 kW head are written down and checked; the CanalPower tests pass and the app runs; the GitHub link (if on your CV) opens to passing code.

## 9 (continued). QGIS — finish the DEM clip and the site map

You already have the right data: Copernicus GLO-30 DEM tiles for the Peshawar/Charsadda corridor (N33_E071, N34_E071, N34_E072), saved locally. Do not mix them with SRTM tiles later. The unfinished step is clipping one tile to your corridor and making a map from it.

1. In QGIS, load tile N34_E072 (the tile your corridor work already used). Draw or finish your corridor polygon and save it as a layer named `corridor_mask`. Save the QGIS project before you clip — an unsaved project is the commonest way this step is "done twice".
2. Clip: Raster → Extraction → Clip Raster by Mask Layer. Input = the DEM tile, mask = `corridor_mask`, output = `dem_clipped.tif` in your project folder. Run it and wait for it to finish; a 40+ MB tile on 8 GB RAM takes time, so do not open other heavy apps during the clip.
3. Verify the result: load `dem_clipped.tif` on its own, check that it covers only the corridor, that its edges are clean, and that its elevation range is plausible for the corridor. Check its properties (CRS, cell size 30 m, file size) and write them in your notes.
4. Build the map layout: clipped DEM as the base, canal lines and your 12 site points on top (only sites whose locations you actually have — do not place a site from a guess), a north arrow, a scale bar, and a legend. Export as PDF and PNG.
5. Write the limit on the map's notes page: a 30 m DEM is a surface model, and it cannot reliably detect a 1–3 m canal fall. Your falls come from field/department data, not from reading small steps off this DEM. Saying that limit aloud is what makes the map credible.

> **Done test — QGIS:** `dem_clipped.tif` exists, opens cleanly, and the exported map shows the corridor, canals, and only the sites you can locate, with a scale bar and legend.

### MATLAB PV — the Simscape model, and the fallback if the licence is missing

1. **Simscape route (licence confirmed, Chapter 3):** open Simulink, add the PV Array block, set its parameters from a real panel datasheet (open-circuit voltage, short-circuit current, and the datasheet values the block asks for — write the datasheet name and panel model in your notes), connect a load/resistor, and run an I–V sweep: vary voltage, record current and power, and plot the I–V and P–V curves. Note the maximum power point from the curve. Save the .slx file and the plots.
2. **Fallback route (no Simscape licence):** build the single-diode model in plain Simulink with basic blocks (current source, diode, series and shunt resistors, as in your earlier guide). Drive it with irradiance and temperature inputs, sweep voltage, and plot the same I–V and P–V curves. This is a complete, honest PV model — present it as "single-diode model built in plain Simulink because Simscape Electrical was not licensed on my machine", not as a Simscape model.
3. Either way, test the model against sense: higher irradiance must raise current and power; higher temperature must lower voltage and maximum power. If your curves do the opposite, the model is wrong — fix the model, do not present the curves.
4. Write one page: model type, panel datasheet (or diode parameters and their source), inputs, the two curves, the maximum power point, and limits (single-diode simplification, no partial shading, no inverter losses unless modelled).

> **Done test — MATLAB PV:** a .slx file that runs, two curves that behave correctly with irradiance and temperature, and a one-page note naming the model type honestly.

## 10. Coursework models — a one-page evidence sheet each

These six projects are finished coursework, but "finished" only helps you if you can answer a question about each one without inventing a number on the spot. Make one sheet per model, using the same template, and keep the sheets in one folder with any photo or model file you still hold.

| Sheet section | What to write (one or two lines each) |
|---|---|
| Project + course | Name, course, semester/year, individual or group (and your part, if group) |
| What it was for | The problem in one sentence |
| What you built / modelled | Physical prototype, MATLAB/Simulink model, or investigation — and the tool used |
| Key inputs | Only numbers you actually used, with units. If you no longer hold the exact numbers, write "inputs not retained — method retained" rather than reconstructing them |
| Key result | What came out, in numbers if you hold them, in words if you do not — labelled either way |
| What you would change | One honest improvement. Professors love this question because it shows you learned, not just completed |
| Evidence held | Photo, model file, report, or "no file retained" — stated plainly |

**The six sheets to write:** (1) Wind turbine prototype — aerodynamic and electromechanical conversion, what the prototype actually generated or demonstrated; (2) DC-DC boost converter — MATLAB model, duty-cycle control, input vs stepped-up output, stand-alone PV context; (3) Sand-based thermal storage — heat retention findings and off-peak dispatch idea, with the limits of a lab-scale investigation stated; (4) Solar-powered electrolyzer — PV input coupled to electrolysis, hydrogen produced or demonstrated, and the safety/rate limits you observed; (5) Fire suppression system — control logic, sensors/actuators in the logic, and the trigger sequence; (6) Concentrated solar water heater — heat-transfer design, what heated how much, and the losses you noted.

> **The rule for all six:** if you do not hold a number any more, say so and talk about the method. A reconstructed "result" that you cannot show is worse than an honest "I no longer hold the raw numbers, but the method was X". A professor can work with the second answer. They cannot work with the first once it unravels.

## 11. The Mitacs interview bank

Professors in matching interviews ask about the CV in front of them. The questions below are the ones your CV invites. Practise them aloud, in your own words, until each answer is under two minutes and ends at a file you can open. Do not memorise these sentences word for word — a memorised answer sounds memorised, and a follow-up question will break it.

**1. "Tell me about your FYP."** "I am assessing low-head hydropower at 12 canal sites in the KP irrigation network. We rank sites physics-first — power from seasonal head, flow, and efficiency — with a model only as a comparative check, and any site that harms irrigation is excluded. The data comes from government departments; [number] sites now have sourced seasonal data, and the ranking sheet shows which cells are still unsourced." Then open the ranking sheet.

**2. "What did you actually do on Energiscope?"** "I built and deployed the platform. I have to be precise about the data: the version you see now is labelled sample data for 2020–2025, and the real NEPRA dataset integration is the next step, with a Methods page stating the source and years. Here is the live site and the Data Note." Never open with the old "10+ years real data" line.

**3. "Is your FYP model trained and validated?"** "No — and I do not claim it is. With 12 sites and no real turbine performance data, a trained-and-validated claim would not be honest. The ranking is physics-based and checkable by hand; the model is a comparative check, and its small-sample and label limits are written in the report."

**4. "Show me your HOMER work."** "Gulabad, Charsadda: an unelectrified site, hourly load from [your load source], solar resource from NASA POWER, and a [PV-hydro-battery / PV-diesel-battery — whichever you ran] system sized in HOMER Pro. The winning system costs [value] per kWh at a renewable fraction of [value]. Here is the .homer file and the report." If the case is not finished, say: "Load and resource are done; the HOMER optimisation runs [dates], and here is the load file."

**5. "What was your part in the floating-PV paper?"** Give your three contribution lines from Chapter 7, then the headline table: floating vs ground mean efficiency, temperature, and output, over how many days. End with: "The manuscript is in preparation; it is not submitted yet."

**6. "What did you do at Tarbela?"** Give the one-page record: placement, supervisor, three personal tasks, operated vs observed. Name the difference between the WAPDA station internship and the PCCCL 5th Extension placement without being asked — it shows you are not blurring them.

## 11 (continued). Tough questions — and what never to claim

**7. "Why should I pick a student whose Python and HOMER are still in progress?"** "Because the progress is checkable: here is a tested power module, a clipped DEM, and a HOMER load file, each with its limits written down. I am not claiming expert level; I am showing the work I have actually finished and the dated plan for the rest." Then show one artefact. One open file beats ten claimed skills.

**8. "What software can you actually use today?"** Answer only with artefacts: Python (script + test), QGIS (clipped DEM + map), MATLAB/Simulink (PV curves or converter model), HOMER Pro (project file or, if unfinished, load + resource files and the trial dates), Excel (floating-PV workbook). Do not list AutoCAD, Energy3D, Helioscope, or Revit as proficiencies unless you can open a file in each — if you cannot, say "used in coursework, not current practice" and move on.

**9. "What will you do in my lab in the first month?"** Tie it to their project, not to yourself: read their two most relevant papers, reproduce one figure or calculation from them, learn their lab's tool from a small test case, and report back with a one-page note. That answer works for geothermal, hybrid systems, storage, or grids — adapt the paper titles to the professor in front of you.

**10. "What are you weakest at?"** Pick a real, current gap with a plan: "My GIS and HOMER work are the newest — the DEM clip and the Gulabad optimisation are dated [dates] in my plan, and here is what each one already produces." A weakness with a dated plan is a strength in an interview. A weakness you deny is a risk.

> **Never claim, in any interview, email, or application:** a publication that is only in preparation; a dataset you have not loaded; a trained/validated model on 12 sites with no turbine data; a HOMER result from a trial you have not run; a licence you do not hold; a site figure with no source; or a skill you cannot demonstrate in the room. If you are unsure whether a claim is safe, use the Chapter 12 test: can I open the file that proves it, right now?

**Before every interview, pack this folder (on the laptop and on a USB drive):** FYP ranking sheet + map, Energiscope screenshots + live link, HOMER file + report (or load/resource files), floating-PV headline table + draft, Tarbela one-pagers, Python test output, QGIS map, MATLAB curves, coursework sheets, and this guide's Chapter 12 checklist. Test that every file opens on the interview machine before the call starts — a file that opens at home and not in the call is not packed.

## 12. Final checklists, folders, and CV updates

### Per-project done checklist (photocopy this page for each project)

[ ] Output file exists and opens  [ ] Inputs and sources written in the file or folder  [ ] Main result explainable in two minutes  [ ] Limits written down  [ ] Screenshots saved  [ ] CV wording updated to match exactly  [ ] Backup copy saved (not only on the laptop disk)

### Folder structure (create once, use for everything)

```
~/projects/
  energiscope/        (P0 screenshots, live-site check, Methods draft)
  fyp-ranking/        (sourced data, power sheet, ranking, map, photos)
  homer-gulabad/      (load CSV, NASA POWER CSV, .homer file, report)
  floating-pv/        (audited workbook copy, headline table, figures, draft)
  tarbela/            (logbook, certificates, one-page summaries)
  skills/             (python-task, qgis-map, matlab-pv, canalpower-repo)
  coursework/         (six evidence sheets + photos/models held)
  interviews/         (Chapter 11 answers practised, pack checklist)
```

### How to update your CV — the only safe rule

Change a CV line only on the day its done test passes, and change it to exactly what the file shows: numbers, dates, and status words copied from the file, not improved in the CV. Keep "in progress" wording for everything else, with no date promise you have not kept before. When a project moves from "in progress" to "completed", keep the old CV version saved, so you can show what changed and when if anyone asks. Never send two different CV versions to the same professor — pick the current one, and resend it whole if it changes.

**One last order of work, if you only have one free evening before an interview:** (1) upload the Energiscope P0 fix, (2) learn the three safe answers in Chapter 1's red box, (3) pack the interview folder with whatever artefacts already exist, (4) write the Tarbela one-pagers. That evening alone removes the biggest risks, even if no other project is finished yet.

> **Final word:** Professors do not expect a final-year student to have finished everything. They expect him to know exactly what is finished, what is not, and what the next step is — and to prove it with files, not adjectives. Finish one project properly, then the next. By the time you have worked through this guide, your CV will not just read better. It will survive being opened.

## Sources and re-check dates

**Your own records used:** refined 2-page CV (Muhammad_Ismail_CV_Refined.pdf, October 2026) and original CV (Muhammad_Ismail_CV.pdf); Energiscope review and P0 fix files (Energiscope_P0_Fixes.zip, 4 November 2026); CanalPower PK starter pack (CanalPower_PK_Starter_Pack.zip, 4 November 2026); FYP roadmap and Copernicus GLO-30 DEM tiles (N33_E071, N34_E071, N34_E072); FYP power module (power_calculator.py) and minimum-head check script; QGIS beginner clipping skill; MATLAB R2025b Simscape Electrical guidance from your November 2026 work.

**Official software sources to re-check on the day you download:** python.org and code.visualstudio.com; qgis.org installation guide (Debian/Ubuntu section, including the Mint codename note); mathworks.com — MATLAB downloads and the Simscape Electrical product page, plus Home → Add-Ons in MATLAB itself; homerenergy.com — HOMER Pro pre-sale FAQ (Windows-based application; 21-day free trial for first-time users) and the download/install/licence article (download from users.homerenergy.com after signing in; run the HomerPro .msi installer); power.larc.nasa.gov (NASA POWER Data Access Viewer); github.com and vercel.com for hosting.

**What to re-check, and when:** HOMER trial length and academic pricing on the day you activate (not the day you read this); Simscape Electrical licence status in your own MATLAB with `ver` and the licence test before you promise a Simscape model; QGIS repository codename for your Mint version at install time; and every 2028-style date in this guide is a plan date, not a deadline set by any university or professor — your interview dates, once given, replace the calendar in Chapter 2.

**What this guide deliberately does not contain:** no licence keys, no passwords (including the old Energiscope admin password, which must be removed, not repeated), no cracked-download links, no invented site figures, no invented HOMER results, and no publication claim for the floating-PV work. If a future version of this guide ever needs a number, it must come from a file you hold, with its source written beside it.

Good luck — and keep the FYP first until 6 November.